Quantum Mechanics II — Exercise Sheet 13

Angular momentum and radial states

Variant: Solutions

Exercise 1 (A cubic harmonic decomposition). In three dimensions let $p(x) = (x_1 + \mathrm{i}x_2)^3$. (i) Show directly that $\Delta p = 0$ and find $L_3 p$. (ii) Identify its orbital spin and weight on the unit sphere. (iii) Then decompose x_3^3 into a homogeneous harmonic polynomial of degree three and r^2 times a harmonic polynomial of degree one.

Solution. The second derivatives in x_1 and x_2 are $6(x_1 + \mathrm{i}x_2)$ and $-6(x_1 + \mathrm{i}x_2)$. The x_3 derivative vanishes, so $\Delta p = 0$. Using $L_3 = -\mathrm{i}\hbar(x_1\partial_2 - x_2\partial_1)$ gives

$$L_3 p = 3\hbar p.$$

The degree-three harmonic space has spin $l = 3$, and this vector has weight $\mu = 3$. Since $\Delta x_3^3 = 6x_3$ and $\Delta(r^2 x_3) = 10x_3$, the decomposition is

$$x_3^3 = \left(x_3^3 - \frac{3}{5}r^2 x_3\right) + r^2\left(\frac{3}{5}x_3\right).$$

Both polynomials inside parentheses are harmonic, of degrees three and one respectively.

Exercise 2 (The first radial excitation of the oscillator). Let $s = \sqrt{\hbar/(m\nu)}$ and $l \geq 0$ be an integer. For the radial oscillator expression

$$H_l = -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}r^2} + \frac{\hbar^2 l(l+1)}{2mr^2} + \frac{m\nu^2 r^2}{2},$$

first verify the energy of $u_0(r) = r^{l+1}e^{-r^2/(2s^2)}$. Then find c and E such that $u_1(r) = r^{l+1}(c - r^2/s^2)e^{-r^2/(2s^2)}$ satisfies $H_l u_1 = E u_1$ for $r > 0$. These are radial expressions; no new self-adjoint realization on the half-line is to be proved.

Solution. Differentiating u_0 gives

$$\frac{u_0''}{u_0} = \frac{l(l+1)}{r^2} - \frac{2l+3}{s^2} + \frac{r^2}{s^4}.$$

The r^{-2} and r^2 terms cancel in $H_l u_0$, leaving $E_0 = \hbar\nu(l + 3/2)$. For $u = u_0 v$ the remaining product rule is

$$H_l(u_0 v) = E_0 u_0 v - \frac{\hbar^2}{2m}u_0\left[v'' + 2\left(\frac{l+1}{r} - \frac{r}{s^2}\right)v'\right].$$

Substituting $v = c - r^2/s^2$ gives the coefficient equations

$$E = \hbar\nu\left(l + \frac{7}{2}\right), \qquad c = l + \frac{3}{2}.$$

The polynomial has one positive radial zero, at $r = s\sqrt{l + 3/2}$.

Exercise 3 (A second radial excitation). Use the radial oscillator expression from Exercise 2, now with $l = 0$. Put $z = r^2/s^2$ and seek a solution

$$u(r) = r\left(z^2 + Az + B\right)e^{-z/2}, \qquad r > 0.$$

(i) Find A, B and the energy E for which $H_0 u = Eu$ as a radial differential identity. (ii) Find the two positive radial zeros and identify the total oscillator excitation N from $E = \hbar\nu(N + 3/2)$. Here H_0 denotes the $l = 0$ radial expression of Exercise 2, not the free Hamiltonian on the line.

Solution. For $u = re^{-z/2}v(z)$, the product rule of Exercise 2 gives

$$H_0 u = \hbar\nu r e^{-z/2}\left[-2zv''(z) + (2z - 3)v'(z) + \frac{3}{2}v(z)\right].$$

Substituting $v = z^2 + Az + B$ gives the polynomial

$$\frac{11}{2}z^2 + \left(\frac{7}{2}A - 10\right)z + \frac{3}{2}B - 3A.$$

Matching its coefficients with $Ev/(\hbar\nu)$ yields

$$E = \frac{11}{2}\hbar\nu, \qquad A = -5, \qquad B = \frac{15}{4}.$$

The two polynomial roots are $(5 - \sqrt{10})/2$ and $(5 + \sqrt{10})/2$, both positive. The positive radial zeros and excitation are

$$r_\pm = s\sqrt{\frac{5 \pm \sqrt{10}}{2}}, \qquad N = 4.$$

This constructs one scalar-sector state at that level without assuming a general radial-family formula.

Exercise 4 (Angular probabilities without a classical orbit). On the unit sphere consider $Y(\theta, \varphi) = C \sin\theta \cos\theta\, e^{i\varphi}$, with $C > 0$. (i) Normalize it with surface measure $\sin\theta\, d\theta\, d\varphi$. (ii) Find the probability of the northern hemisphere and $\langle\cos^2\theta\rangle$. (iii) Identify l and μ using the harmonic polynomial $x_3(x_1 + ix_2)$.

Solution. With $z = \cos\theta$, the norm integral is $2\pi C^2 \int_{-1}^{1}(z^2 - z^4)\, dz$. Thus

$$C = \sqrt{\frac{15}{8\pi}}, \qquad \mathrm{Prob}(x_3 > 0) = \frac{1}{2}.$$

The cosine second moment is the ratio of two polynomial integrals:

$$\langle\cos^2\theta\rangle = \frac{\int_{-1}^{1}(z^4 - z^6)\, dz}{\int_{-1}^{1}(z^2 - z^4)\, dz} = \frac{3}{7}.$$

The polynomial is harmonic, homogeneous of degree two, and has L_3 weight $\hbar$. Therefore $l = 2$ and $\mu = 1$. These are angular probabilities of a stationary angular state, not positions along a classical orbit.